AEWFNet: Adaptive Enhancement and Wavelet Convolution for Hyperspectral and Multispectral Image Fusion

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***Abstract*—Hyperspectral and multispectral image fusion (HMIF) represents a highly effective approach to enhancing the spatial resolution of hyperspectral images (HSIs). However, due to the inaccurate modeling of the spectral response function (SRF) and limited capability in capturing high-frequency details, many existing methods still struggle to maintain spectral consistency and effectively represent spatial structures. To address the aforementioned issues, this study proposes an Adaptive Enhancement and Wavelet Convolution-based HMIF framework, termed AEWFNet, which aims to improve the spatial structural representation and spectral reconstruction accuracy of the fused images. First, a spatial-aware response learning (SARL) mechanism is proposed to structurally optimize and guide the spectral degradation learning module, helping the model better understand the complex interactions between spatial structures and spectral information, thereby improving the fitting accuracy to the real degradation process. Second, an attention-based rotation prediction augmentation network (ARPAN) is designed to optimize spectral information through an adaptive enhancement strategy, effectively improving the generalization performance and feature representation of images under various rotation conditions. Finally, wavelet convolution is employed to enhance the spatial feature representation across multiple scales. Experimental results on several datasets show that AEWFNet outperforms the compared methods in both quantitative and qualitative evaluations, achieving better preservation of the spatial–spectral characteristics of the images. The code for AEWFNet will be made publicly available at https://github.com/LRuiRui517/AEWFNet**

***Index Terms*—AEWFNet, attention-based rotation prediction augmentation network (ARPAN), hyperspectral and multispectral image fusion (HMIF), spatial-aware response learning (SARL).**

I. Introduction

HYPERSPECTRAL images (HSIs), composed of tens or even hundreds of narrow spectral bands, provide more detailed spectral information, enabling the precise identification of different surface materials [1]. However, due to the physical limitations of imaging sensors, there exists a trade-off between spatial and spectral resolution, leading to the typically low spatial resolution of HSIs and thereby restricting their range of applications. Multispectral images (MSIs), which contain only a limited number of broad spectral bands, have lower spectral resolution compared to HSIs but offer higher spatial resolution, effectively complementing the fine-grained perception limitations of HSIs [2]. Against this background, hyperspectral and multispectral image fusion (HMIF) has emerged as an effective approach to enhance the spatial resolution of HSIs by combining the spectral advantages of HSIs with the spatial advantages of MSIs [3], [4]. The fused images can significantly improve the reliability of downstream tasks such as target recognition, land cover classification, and environmental monitoring [5], [6].

Existing methods for the HMIF problem can be broadly categorized into two types: model-based approaches [7] and deep learning-based approaches [8]. The former relies on imaging degradation models and establishes a physical mapping between low spatial resolution HSIs (LR-HSI) and high spatial resolution MSIs (HR-MSI). In this framework, the fusion task is formulated as an ill-posed inverse problem, where regularization techniques are introduced to constrain the solution space and enable the reconstruction of the high-resolution HSI (HR-HSI) [9], [10]. These methods offer clear theoretical foundations and strong interpretability; however, their practical application is often limited by inaccurate degradation model assumptions and the complexity of parameter tuning.

In contrast, deep learning-based fusion methods leverage deep neural networks to model the cross-modal correspondence between HSI and MSI in an end-to-end manner, enabling nonlinear feature alignment and enhanced reconstruction performance through their powerful representation capabilities [11]. However, the limitation imposed on model training by the scarcity of hyperspectral data still persists, as insufficient data make it difficult for models to fully learn the spectral diversity characteristics present in real-world scenarios [12]. To alleviate the aforementioned challenges, researchers have increasingly explored unsupervised deep fusion methods, aiming to achieve effective model training and improve cross-domain generalization capability in the absence of real HR-HSI labels [13], [14], [15]. Although these methods have made certain progress in the HMIF task, there are still significant shortcomings in physical modeling during the fusion process and the coordination of multisource information. Most existing approaches model the spectral

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response function (SRF) in the spectral degradation process rather coarsely, overlooking the heterogeneity across shadowed, highlight, smooth, and highly textured regions in real scenes. This simplification often leads to oversmoothing, color shifts, and local detail loss, fails to faithfully reflect the sensor's physical imaging mechanism, and ultimately undermines the fidelity and interpretability of the fusion results. Moreover, in spatial–spectral joint modeling, there remains a lack of effective mechanisms to exploit the complementary characteristics of HSI and MSI, which frequently causes missing details in spatial structure recovery and insufficient cross-scale representational capacity. These problems significantly limit the adaptability and generalization ability of existing models in complex real-world scenarios, representing core challenges that need to be addressed in current HMIF research.

Inspired by recent progress in HMIF, this study proposes an HMIF approach based on adaptive augmentation and wavelet convolution. The method comprises three stages that together form a phased fusion framework. First, we design a spatial-aware response learning (SARL) mechanism that injects spatial context to dynamically model spectral degradation, thereby generating training pairs that more faithfully reflect the real imaging process. Second, unlike random geometric augmentation, we elevate augmentation from passive sampling to active selection via an attention-based rotation prediction augmentation network (ARPAN), which predicts sample-specific rotation angles and applies geometric transforms to strengthen spatial–spectral representations, improving robustness and generalization to geometric variations. Finally, we construct the fusion backbone with an interpretable decompose-then-reconstruct pipeline: an efficient upsampling module that couples discrete wavelet transform (WT) with convolution enables fine recovery of multiscale spatial structures and spectral details. The main contributions of this study can be summarized as follows.

1) To address the issue of traditional spectral degradation modeling relying on fixed SRF and its inability to adapt to complex scenarios, we introduce spatial-aware information to dynamically adjust degradation kernel parameters, enabling adaptive modeling of the spectral degradation process. This effectively improves the model's ability to fit real imaging mechanisms and enhances regional robustness.
2) We propose the ARPAN, designed to enhance the representation of spatial and spectral features and to tackle challenges arising from rotations and scale variations in HSIs. By adaptively predicting rotation angles and performing the corresponding geometric transformations, ARPAN improves fusion effectiveness and strengthens the model's generalization under diverse conditions.
3) To address the limitation of traditional upsampling methods in recovering high-frequency details, we integrate WT with convolution operations to extract multiscale features and accurately reconstruct high-frequency information, thereby improving spatial structure fidelity while enhancing spectral consistency.
4) Achieved Comprehensive Performance Leadership Across Multiple Datasets and Scaling Factors: AEWFNet significantly outperforms existing methods on four publicly available HSI fusion datasets, including PaviaC, achieving state-of-the-art (SOTA) performance across multiple metrics. This fully validates the effectiveness and generalizability of the proposed method in spatial detail reconstruction and spectral fidelity.

The remainder of this study is organized as follows. Section II of this study presents the related work. Section III details the network architecture of AEWFNet and the associated loss functions. Section IV presents the experimental results of AEWFNet and comparative methods on four publicly available HSI datasets, along with ablation studies. Section V concludes the study and outlines directions for future work.

II. Related Work

A. Model-Based HMIF

In order to enhance the fusion quality and stability of HR-HSI, researchers frequently introduce various forms of prior knowledge as constraints [16]. Among the numerous existing approaches, matrix factorization [17] and tensor modeling [18] have become mainstream strategies, modeling and characterizing the underlying structure of HR-HSI from different dimensional perspectives. Matrix factorization treats HSIs as 2-D matrices and performs low-rank modeling from either the spectral or spatial perspective. For instance, Lin et al. [19] proposed a rapid fusion framework based on truncated matrix singular value decomposition (TMSVD), which transforms the fusion problem into the estimation of three factor matrices by exploiting the similarity between the HR-HSI and HR-MSI after TMSVD. Shen et al. [20] transformed the fusion problem into spectral optimization and spatial optimization problems using matrix decomposition. The initial HR-HSI image was obtained by solving a system of linear equations and subsequently multiplying the solutions of both. Yang et al. [21] employed compressed sensing techniques to randomly sample the low-rank background signals from the observed data and constrained the search space by projecting endmember vectors onto the orthogonal subspace of the low-rank background, thereby preventing disordered expansion of the solution space. Although such methods offer certain advantages in handling 2-D structures, they are relatively limited in their ability to represent the coupled spatial–spectral information inherent in the 3-D structure of HSIs.

Considering that HSI inherently possesses a 3-D or even higher dimensional structure, tensor modeling methods approach the problem from the perspective of multidimensional collaborative modeling, capturing the spatial–spectral coupling relationship more comprehensively. For instance, Xu et al. [22] imposed an L_1 norm constraint on the nuclear tensor in order to characterize sparsity, while applying a unidirectional total variation (TV) regularization term on three dictionary matrices to model local smooth structures in both spatial and spectral directions. Zheng et al. [23] proposed a

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unified cloud-based HSI–MSI fusion framework based on the general distributed alternating direction method of multipliers (ADMM) and introduced an adaptive CP rank estimation method to achieve better super-resolution reconstruction performance. Xu et al. [24] proposed an HMIF method based on coupled tensor double factor (CTDF). In this approach, the HR-HSI is represented by a fourth-order spatial factor and a third-order spectral factor, which are connected through tensor contraction. However, tensor-based modeling methods generally suffer from complex model structures, difficulties in parameter estimation, and high computational cost, which pose significant challenges for practical applications. Such model-based approaches demonstrate strong advantages in reconstruction accuracy, their performance heavily depends on the rationality of the prior model design. Since different application scenarios exhibit varying image characteristics, it is often necessary to construct specific prior models tailored to individual cases [25].

B. Deep Learning-Based HMIF

According to the availability of supervision signals, existing deep learning-based HMIF methods can be broadly categorized into supervised [26] and unsupervised [27] approaches. Supervised learning methods, by constructing end-to-end deep networks, learn the complex cross-modal mapping between paired LR-HSI and HR-MSI datasets. In recent years, significant progress has been made in terms of fusion quality. For example, Ma et al. [28] proposed a fusion strategy based on a dual cross transformer (DCT), which successfully captures the interactions between HSI and MSI through a directional paired multihead cross-attention mechanism. Hsu et al. [29] employed a dual-stream network structure (DTS) during the fusion process, extracting both joint and independent features of LR-HSI and HR-MSI, thereby improving spatial and spectral quality while reducing computational complexity. Yan et al. [30] proposed a spatial–spectral unfolding network with mutual guidance (SMGU-Net), which incorporates a mutual guidance mechanism and is optimized using a semiquadratic splitting and gradient descent algorithm. This framework is unfolded into a neural network structure, effectively enhancing the quality and fidelity of the fused images. Zhu et al. [31] combined the long-range modeling advantage of Mamba with the capability of implicit neural representation (INR) in continuous domain processing, and their constructed MCIFNet demonstrated excellent performance in spatial and spectral detail reconstruction. Despite the excellent performance of supervised methods, their training process typically relies on high-quality paired data samples, which are difficult to obtain widely in hyperspectral remote sensing tasks, limiting the model's generalization and deployment capabilities.

In order to address the scarcity of supervised data, researchers have increasingly turned to unsupervised fusion frameworks. These approaches leverage data self-expression, self-reconstruction, or auxiliary degradation models to enable model training without relying on ground-truth HR-HSI labels, thereby improving fusion performance and enhancing adaptability in practical applications. For example, Cao et al. [32] proposed an unsupervised hybrid network of transformer and CNN (uHNT), which effectively captures the interaction information between the spatial and spectral domains through a multilevel cross-feature attention mechanism, and avoids reliance on known degradation information by employing unsupervised joint training. Sun et al. [33] proposed an unsupervised 3-D tensor subspace decomposition network, which leverages a 3-D tensor subspace extraction and reconstruction framework to effectively predict and reconstruct high spatial resolution HSIs. Gao et al. [34] proposed the enhanced unmixing-inspired unsupervised network with attention-embedded degradation learning (EU2ADL) model, which comprehensively extracts information interactions through coupled autoencoders and introduces a novel hybrid loss function to enhance spatial–spectral representation capability. Zheng et al. [35] proposed a fusion method based on an unsupervised pretraining framework (UPFW), which consists of three stages: first, the model learns the spatial and spectral degradation parameters of the input images; second, a supervised pretraining is conducted at low resolution using an adaptive fusion network (ADFNet); and finally, the pretrained model is applied for unsupervised fusion at high resolution to enhance generalization and practical adaptability. Overall, due to their low dependence on labeled data, greater flexibility in degradation assumptions, and strong cross-domain generalization capability, unsupervised methods are emerging as a key development direction in HMIF research.

C. Degradation Model

In existing supervised HMIF methods, a widely adopted approach is to construct solutions based on a degradation model [36], [37]. The model typically assumes the HR-HSI as the target to be reconstructed, while the HR-MSI and LR-HSI are regarded as its degraded observations in the spatial and spectral dimensions, respectively. Specifically, spatial degradation is commonly implemented through convolution operations associated with the point spread functions (PSF) followed by downsampling, whereas spectral degradation is achieved by applying the SRF to perform a weighted integration of the HR-HSI, resulting in the MSI observation.

Under this modeling framework, HMIF is formalized as an inverse problem: given the HR-MSI and LR-HSI, the goal is to jointly estimate the latent HR-HSI. In recent years, various methods have been proposed to enhance the expressiveness and adaptability of the degradation modeling. For instance, Qin et al. [14] employed a single convolutional layer to simulate the SRF, integrating the spectral dimensionality reduction process into end-to-end network training, thereby effectively enhancing the flexibility of degradation modeling. Zheng et al. [35] represented the HSI within an endmember–abundance decomposition framework as the product of an endmember matrix and an abundance matrix, and introduced an SRF matrix to perform dimensionality reduction on the endmember matrix, thus simulating the spectral degradation of practical multispectral imaging systems. Jia et al. [38] introduced a spatial degradation matrix and a spectral degradation matrix to separately model the degradation processes of HSI and MSI, respectively. During the pretraining phase, two

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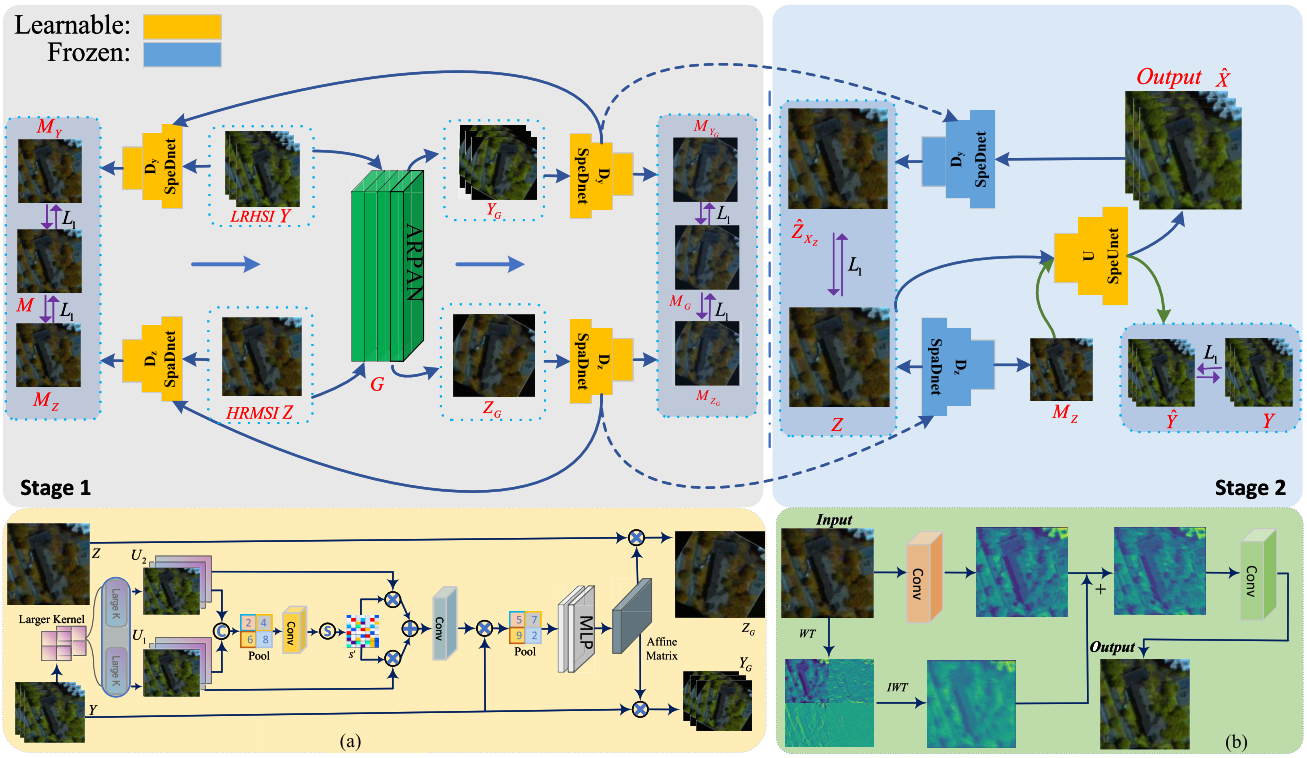


Fig. 1. Overall framework diagram. (a) ARPAN. (b) SpeUnet.

upsampling modules were jointly learned, effectively promoting the complementary fusion of HSI spectral details and MSI spatial details, thereby enhancing the spectral–spatial consistency and fine-grained information recovery capability of the final fused image.

Although the aforementioned methods have made notable progress in degradation modeling and fusion performance, there are still significant variations in spectral response characteristics across different land cover categories, texture structures, and spatial scales in real-world imaging scenarios. Relying on a unified and static degradation modeling strategy is insufficient to capture the complex structural variations among regions, often resulting in local distortions or oversmoothing during fine-grained spectral reconstruction and spatial detail recovery. Therefore, it is essential to develop a method that can dynamically adjust the degradation modeling strategy based on local image features. This will enable the model to adapt more effectively to spatialspectral variations across regions, enhancing its adaptability and robustness in complex real-world scenarios.

III. Method

In this section, this study provides a detailed introduction to a novel HMIF method, which is based on a deep learning framework and incorporates a physical degradation model to improve fusion performance. First, an overview of the overall framework is presented, followed by detailed descriptions of each submodule, including the Spectral Downsampling network (SpeDnet), Spatial Downsampling network (SpaDnet), ARPAN, and the technical details of the Spectral Upsampling network (SpeUnet), which employs wavelet convolution for efficient spectral upsampling.

A. Overall Framework

As demonstrated in Fig. 1, the proposed AEWFNet comprises four fundamental modules: degradation modeling, adaptive enhancement, feature fusion, and detail restoration. Collectively, these modules constitute a comprehensive HMIF pipeline. First, a spectral downsampling network D_y and a spatial downsampling network D_z are designed to simulate the spectral and spatial degradation processes in real-world imaging. To enhance the modeling capability for complex regional characteristics, an SARL mechanism is introduced to dynamically adjust the spectral degradation kernel based on local spatial structures of the image. Next, the ARPAN module adaptively predicts the rotation angle of the image and applies corresponding geometric transformations, effectively improving the model's adaptability to spatial perturbations and its feature representation capacity. The feature fusion module, which serves as the core of the backbone network, enables joint modeling and cross-modal fusion of LR-HSI and HR-MSI features. The detail restoration module builds on this foundation, utilizing a combination of WT and convolutional operations for upsampling. This approach serves to further enhance spatial detail and spectral consistency in the fused image. To improve training stability and clarify functional responsibilities, each module is configured in either a learnable or frozen state during different training stages. The frozen state is used to fix the parameters learned in earlier stages, ensuring structural decoupling between degradation modeling and fusion reconstruction, while maintaining functional synergy—thereby facilitating the overall optimization of fusion performance [39]. The detailed technical description and analysis of each module are provided in Sections III-B–III-G.

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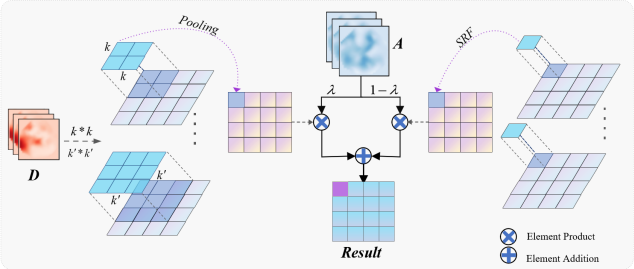


Fig. 2. SARL mechanism.

B. Degradation Operation

To achieve high-quality HMIF, it is essential to first model the degradation relationships among different observed images. Let $X \in \mathbb{R}^{WH\times C}$ represent the HR-HSI considered as the fusion target, and $Y \in \mathbb{R}^{wh\times C}$ and $Z \in \mathbb{R}^{WH\times C_m}$ represent the LR-HSI and HR-MSI, which are the spatially degraded and spectrally degraded versions of X, respectively. Here, W, H, and C represent the width, height, and number of spectral bands of X, respectively; w and h denote the width and height of Y, and C_m is the number of spectral bands in Z, with $w \ll W$, $h \ll H$, and $C_m \ll C$ typically holding. These degradation models can be expressed as follows:

$$Y = \mathrm{PX} \tag{1}$$

$$Z = \mathrm{XS} \tag{2}$$

where $P \in \mathbb{R}^{wh\times WH}$ denotes the spatial degradation operator, which is typically composed of a PSF-related blurring matrix and a spatial downsampling operation. $S \in \mathbb{R}^{C\times C_m}$ represents the spectral downsampling operator associated with the SRF. Based on the degradation model described above, the objective of the HMIF task is to reconstruct the latent HR-HSI X from the spatially and spectrally degraded observations Y and Z, respectively. Additionally, according to the properties of the degradation model, the spectral degradation result M_Y of Y should be equivalent to the spatial degradation result M_Z of Z, i.e.,

$$M_Y = \mathrm{YS} = \mathrm{PZ} = M_Z \tag{3}$$

where $M_Y \in \mathbb{R}^{wh\times C_m}$ and $M_Z \in \mathbb{R}^{wh\times C_m}$ represent the spectrally degraded HSI and the spatially degraded MSI, respectively. This degradation model provides a theoretical foundation for the subsequent fusion network design, ensuring the effective reconstruction of the latent high-resolution HSI from various degraded observations.

C. Spectral Downsampling Network

The SpeDnet is a core component used to simulate the spectral degradation process during image acquisition. The aim of this module is to downsample the HSI based on the spectral response characteristics of the imaging sensor. This generates the corresponding HR-MSI, which is then used as one of the inputs to the fusion module [40], [41]. In conventional methods, the SRF is usually modeled using static parameters. Building upon this, we implement the downsampling operation as a 1×1 convolution, assigning a learnable response weight to each spectral channel to simulate sensor-specific spectral sensitivity

$$y_{\mathrm{SRF}} = \sum_{i=1}^{C_{\mathrm{in}}} W_i * x_i \tag{4}$$

where W_i denotes the convolution kernel of the SRF, x_i is the input image, and * represents the convolution operation, where the output y_{SRF} represents the result obtained using static SRF modeling. However, this fixed matrix design does not account for variations in spectral responses across different sensors and imaging conditions. It is also unable to effectively capture the nonlinear and dynamic characteristics frequently observed in real-world imaging scenarios. To improve the flexibility and adaptability of spectral modeling, we propose an SARL mechanism that incorporates spatial contextual information to guide the SRF modeling process. The spatial information is represented as follows:

$$y = \sum_{j=1}^{C_{\mathrm{in}}} W_{ij} * x_j \tag{5}$$

where W_{ij} denotes the weight at the ith row and jth column of the convolution kernel, and x_j is the jth pixel value of the input image. A global average pooling operation is utilized to extract the global features of the input image

$$F_{i,j} = \frac{1}{k\times k}\sum_{m=0}^{k-1}\sum_{n=0}^{k-1} P_{i+m,j+n}. \tag{6}$$

Each element $F_{i,j}$ of the output feature map represents the average value within the pooling window, where $P_{i+m,j+n}$ denotes the pixel value of the input feature map, and $k \times k$ is the size of the pooling window. Subsequently, a multilayer perceptron (MLP) consisting of two fully connected layers and a ReLU activation function is employed to adaptively generate convolution kernel parameters based on spatial features. These dynamically generated kernels are then used to convolve the input image, producing the dynamic spectral response output y_{dynamic}

$$y_{\mathrm{dynamic}} = \textstyle\sum_{j=1}^{C_{\mathrm{in}}} W_{\mathrm{ij}} * x_j + b_i. \tag{7}$$

The module D in Fig. 2 represents the generated adaptive convolution kernel, and b_i denotes the bias term, which corresponds to a constant added to each element of the output image to adjust the convolution result. The final output y_{final} is constructed by fusing the output from the static SRF and the dynamically generated convolution result

$$y_{\mathrm{final}} = \lambda \cdot y_{\mathrm{dynamic}} + (1-\lambda)\cdot y_{\mathrm{SRF}} \tag{8}$$

where λ denotes the fusion weight, which is generated by an auxiliary convolutional network followed by a Sigmoid activation function [42], [43]. Its value ranges from 0 to 1, representing the fusion ratio between the dynamically and statically modeled outputs. As illustrated in Fig. 2, this mechanism effectively enhances the adaptability of the SRF to diverse regional features. Finally, in the spectral downsampling process, the input image $Y \in \mathbb{R}^{\mathrm{wh}\times C}$ is processed by module

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D_y to generate the corresponding HR-MSI $M_Y \in \mathbb{R}^{\mathrm{wh}\times C_m}$. The corresponding modeling expression is given as follows:

$$M_Y = \mathrm{SpeDnet}(Y) = \sum_{i=1}^{C_{in}} \left[\lambda_{\mathrm{adaptive}} \cdot (K_i * Y_i) + \left(1-\lambda_{\mathrm{adaptive}}\right)\cdot (W_i * Y_i)\right] \quad (9)$$

where K_i denotes the adaptively generated convolution kernel, W_i represents the static SRF kernel, and $\lambda_{\mathrm{adaptive}}$ is the fusion weight. This design retains an interpretable static component while introducing a dynamic, locally structure-sensitive component. A scalar gating mechanism adaptively modulates their fusion strength, thereby preventing both overfusion and underfusion.

D. Spatial Downsampling Network

To accurately model the spatial degradation phenomena in real image acquisition, this study constructs a spatial downsampling network D_z to simulate the image degradation process. Specifically, a convolution kernel of size $1 \times r \times r$ with a stride of r is applied to each band along the spatial dimensions, where $r(r = W/w = H/h)$ denotes both the spatial scaling factor and the kernel size. The spatial downsampling network is formulated as follows:

$$M_Z = \mathrm{SpaDnet}(Z) = \sum_{i=1}^{C_{\mathrm{in}}} \left[K_i' * Z_i\right] \quad (10)$$

where M_Z is the result after spatial downsampling. Z denotes the HR-MSI. K_i' denotes the convolution kernel, Z_i represents the ith channel of the input, and C_{in} is the number of channels in the input image Z.

E. Attention-Based Rotation Prediction Augmentation Network

To enhance the model's capacity to handle HSIs under rotation and scale variations, we replace random geometric perturbations with a content-driven selection of the most effective rotation angle, thereby continually mining hard examples and improving worst-case performance. Accordingly, we propose an ARPAN, as illustrated in Fig. 1(a). The core idea of ARPAN is to adaptively extract multiscale spatial and spectral features using large-kernel convolutions, enabling the prediction of the rotation angle of the input image. This predicted angle then guides an affine transformation of the image, ensuring geometric consistency during the augmentation process. While convolutional neural networks are somewhat robust to minor rotations, traditional fixed-angle or random augmentation strategies frequently fail to maintain spatial consistency across spectral bands in HMIF tasks. This is largely due to the high spectral dimensionality and complex spatial structures inherent in HSIs, which can result in spectral distortion and unstable training [44]. Most existing rotation-based methods are designed for natural images, overlooking the spatial–spectral coupling inherent in HSIs, and typically apply a uniform global transformation, lacking the ability to model local variations. To address these issues, ARPAN integrates both spatial and spectral feature information to adaptively predict rotation angles based on image content. This approach provides a framework for guiding the geometric transformation during augmentation, with the objective of achieving better alignment with local structural characteristics. The implementation of this framework has the potential to enhance sample diversity and improve structural alignment under rotational variations. The module has been shown to effectively alleviate spatial–spectral distortion during augmentation, enhancing the model's generalization ability and representational stability under varying pose conditions.

Large-kernel convolutions are employed to capture richer cross-band spectral features and spatial structural information. The feature maps extracted by convolution kernels of different scales are concatenated to obtain a multiscale feature representation

$$U = [U_1; U_2; \ldots U_i]. \quad (11)$$

To extract spectral–spatial relationships, channelwise average pooling [denoted as $P_{\mathrm{avg}}(\cdot)$] and max pooling [denoted as $P_{\mathrm{max}}(\cdot)$] operations are applied

$$\mathrm{SA}_{\mathrm{avg}} = P_{\mathrm{avg}}(U) \quad (12)$$

$$\mathrm{SA}_{\mathrm{max}} = P_{\mathrm{max}}(U) \quad (13)$$

where $\mathrm{SA}_{\mathrm{avg}}$ and $\mathrm{SA}_{\mathrm{max}}$ are spatial feature descriptors extracted through average pooling and max pooling, respectively. To facilitate information interaction between different spectral–spatial feature descriptors, the pooled features are concatenated and passed through a convolution layer $F_{2\to N}(\cdot)$ to generate N spatial attention maps

$$\mathrm{SA} = F_{2\to N}\left(\left[\mathrm{SA}_{\mathrm{avg}}; \mathrm{SA}_{\mathrm{max}}\right]\right). \quad (14)$$

For each spatial attention map SA_i, the Sigmoid activation function is applied to obtain the individual spatial selection mask for each decomposed large convolution kernel

$$\mathrm{SA}_i = \sigma(\mathrm{SA}_i) \quad (15)$$

where $\sigma(\cdot)$ denotes the Sigmoid function. Next, the corresponding spectral–spatial selection masks are used to perform weighted fusion of the spectral feature maps extracted by convolution kernels of different scales, and the fused features are passed through the convolution layer $F(\cdot)$ to obtain the attention features

$$S' = F\left(\sum_{i=1}^{N}(\mathrm{SA}_i \cdot U_i)\right). \quad (16)$$

The input image Q and S' are multiplied elementwise to obtain the output Y' for spectral feature extraction and enhancement

$$Y' = Q \cdot S'. \quad (17)$$

Building upon the extracted and enhanced spectral features, the next step is to utilize these enhanced features for rotation angle prediction and image augmentation. Specifically, global average pooling is applied to the enhanced feature map Y'

$$Y_{\mathrm{pool}}' = P_{\mathrm{avg}}\left(Y'\right). \quad (18)$$

The pooled result $Y_{\mathrm{pool}}' \in \mathbb{R}^{B\times C}$ is obtained, which is then passed through two fully connected layers, fc1 and fc2, to

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perform the mapping and ultimately predict the rotation angle $\theta_{\text{pred}} \in \mathbb{R}^B$

$$\theta_{\text{pred}} = \tanh\left(\text{fc2}\left(\text{ReLU}\left(\text{fc1}\left(Y_{\text{pool}}'\right)\right)\right)\right) \times \pi. \tag{19}$$

Based on the predicted rotation angle θ_{pred}, the original input image Q is adaptively rotated. Rotation augmentation helps the network better adapt to different image angles, thereby maintaining the consistency of spectral information and enhancing the network's spatial perception ability across multiple angles. The rotation operation is implemented using an affine transformation matrix, where the affine transformation matrix $\widehat{\theta}$ is defined as follows:

$$\widehat{\theta} = \begin{bmatrix} \cos\left(\theta_{\text{pred}}\right) & -\sin\left(\theta_{\text{pred}}\right) \\ \sin\left(\theta_{\text{pred}}\right) & \cos\left(\theta_{\text{pred}}\right) \end{bmatrix}. \tag{20}$$

The rotation matrix $\widehat{\theta}$ is applied to perform spatial transformation on the image, resulting in the rotated image Q_{rotated}, that is,

$$Q_{\text{rotated}} = \text{Rotate}\left(Q, \hat{\theta}\right) \tag{21}$$

where $\text{Rotate}(Q, \hat{\theta})$ denotes the rotated image obtained by applying the affine transformation matrix $\hat{\theta}$ to the input image Q [45]. When the inputs are Y, Z, and the low spatial resolution version M provided in the real dataset, the corresponding outputs are Y_G, Z_G, and M_G, respectively. This design enables the model to adaptively align spatial structures under varying orientations, enhancing feature consistency and improving the reliability of subsequent fusion stages.

F. Spectral Upsampling Network

Traditional detail injection methods typically employ multiresolution decomposition techniques to extract missing high-frequency details from low spatial resolution images and inject them into the low-resolution images to enhance their resolution [46]. However, these approaches often rely on simple high-frequency extraction and direct injection, which may lead to incomplete detail recovery or the introduction of artifacts when dealing with complex image structures and nonlinear degradations. In the SpeUnet architecture shown in Fig. 1(b), WT not only effectively separates the low-frequency and high-frequency information of the image during the detail restoration process, but also enhances the high-frequency features of the image by combining the high-frequency details with the output of the convolutional layers. This approach avoids the loss of details caused by standard convolution operations, thereby improving the visual quality of the image and the performance of downstream tasks [47].

By combining operations along both spatial dimensions of the input image, a depthwise convolution with a stride of 2 is performed, resulting in each channel having half the spatial resolution of the original image. This process is implemented using the following four wavelet filters [48], [49]:

$$f_{\text{LL}} = \frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \quad f_{\text{LH}} = \frac{1}{2}\begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}$$
$$f_{\text{HL}} = \frac{1}{2}\begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}, \quad f_{\text{HH}} = \frac{1}{2}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \tag{22}$$

where f_{LL} is the low-pass filter, and f_{LH}, f_{HL}, and f_{HH} are a set of high-pass filters. These filters are used to extract the low-frequency component Q_{LL} and the high-frequency components Q_{LH}, Q_{HL}, and Q_{HH} of the spectral image, respectively. The low-frequency component contains the overall structure of the spectral image, while the high-frequency components capture detailed information such as edges, textures, and spectral variations. Through this decomposition, the low- and high-frequency information of the image is effectively separated, allowing for targeted processing and enhancement. The filters in (22) form an orthogonal basis, and the IWT can be implemented using transposed convolution

$$Q = \text{Conv} - \text{transposed}\left(\left[f_{\text{LL}}, f_{\text{LH}}, f_{\text{HL}}, f_{\text{HH}}\right]\right.$$
$$\left.\left[Q_{\text{LL}}, Q_{\text{LH}}, Q_{\text{HL}}, Q_{\text{HH}}\right]\right). \tag{23}$$

The image processed by wavelet convolution, denoted as Q^{wavelet}, can be represented as follows:

$$Q^{\text{wavelet}} = \text{Conv}\left(w, Q\right) + \text{IWT}\left(\text{Conv}\left(w, \text{WT}\left(Q\right)\right)\right) \tag{24}$$

where w denotes the weight tensor of a 3×3 depthwise convolution kernel. After wavelet convolution, a 1×1 convolution layer is applied to map the output features to the target number of HSI channels. This convolution operation preserves spatial information while generating the final HSI. This process can be formulated as follows:

$$Q' = \text{Conv2d}\left(W_1, Q^{\text{wavelet}}\right) + b_1 \tag{25}$$

where Q' denotes the spectral upsampling result, W_1 is the 1×1 convolution kernel, and b_1 is the bias term. When the input to SpeUnet is Z, the output corresponds to the desired HR-HSI. By combining wavelet-based frequency decomposition with convolution-based reconstruction, this design effectively preserves spectral–spatial structures essential for HR-HSI.

G. Loss Function Design

1) Downsampling Network Learning Strategy: To enhance the learning capability of the network and generate more challenging samples, this study employs a data augmenter G and two downsampling networks, D_y and D_z, for training. The entire process is optimized through iterative training. A tunable parameter α is set, and the consistency among M_{Y_G}, M_G, and M_{Y_G} is enforced by minimizing the following loss function:

$$L_G = \alpha L\left(M_{Y_G}\right) + \alpha L\left(M_{Z_G}\right) \tag{26}$$

$$L\left(M_{Y_G}\right) = \log\left(\frac{1}{\text{wh}C_m}\|M_{Y_G} - M_G\|_1\right) \tag{27}$$

$$L\left(M_{Z_G}\right) = \log\left(\frac{1}{\text{wh}C_m}\|M_{Z_G} - M_G\|_1\right) \tag{28}$$

where $\|\cdot\|$ denotes the L_1 norm. During the training of the downsampling networks D_z and D_y, the augmenter G is fixed. The original image Y and its augmented version Y_G are input into D_y, producing M_Y and M_{Y_G}; similarly, Z and its augmented version Z_G are input into D_z, resulting in M_Z and M_{Z_G}. Finally, these two downsampling networks are optimized by minimizing the L_D loss function

$$L_D = \|M_Y - M\|_1 + \|M_{Y_G} - M_G\|_1$$

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TABLE I
PARAMETERS OF THE FOUR DATASETS

Datasets	PaviaC	Houston18	PaviaU	Chikusei
HSI bands	102	46	93	110
MSI bands	4	8	4	8
Spatial resolution (m)	1.3	1	1.3	2.5
Spectral range (nm)	430-860	380-1050	430-860	363-1018
Size	1096×1096	1202×4172	610×340	2517×2335

$$+\|M_Z - M\|_1 + \|M_{Z_G} - M_G\|_1 \tag{29}$$

where M denotes the low spatial resolution version provided in the ground-truth dataset. Through this training strategy, G and the downsampling networks can work collaboratively to improve the quality and consistency of HSI and MSI during the augmentation and downsampling processes.

2) Upsampling Learning Strategy: According to (2) and (3), M_Z and Z are obtained by applying the same spectral degradation operation S to Y and X, respectively. Therefore, by applying the spectral inverse mapping from M_Z to Y, the potential X can be recovered from Z, and this process can be represented as follows:

$$\hat{Y}_{M_Z} = \text{SpeUnet}\,(M_Z) \tag{30}$$

where $\hat{Y}_{M_Z}$ denotes the output of SpeUnet. SpeUnet is optimized by minimizing the loss function L_{S1}

$$L_{S1} = \frac{1}{\text{whC}}\|Y - \hat{Y}_{M_Z}\|_1. \tag{31}$$

After appropriate optimization, Z can be input into SpeUnet for a coarse reconstruction of X. However, the learned low-resolution spectral inverse mapping exhibits certain performance limitations. To achieve a more precise reconstruction of X, this study further utilizes paired Z and X as training data to train SpeUnet. While maintaining L_{S1}, the loss L_{S2} between $\hat{Z}_{X_Z}$ and Z is minimized to optimize both global image consistency and local structural details, formulated as follows:

$$\hat{Z}_{X_Z} = \text{SpeDnet}\left(\text{SpeUnet}\,(Z)\right) \tag{32}$$

$$L_{S2} = \frac{1}{\text{WHC}_m}\|Z - \hat{Z}_{X_Z}\|_1. \tag{33}$$

Finally, SpeUnet is optimized by minimizing the loss function L_S

$$L_S = L_{S1} + \beta L_{S2} \tag{34}$$

where β is a tunable hyperparameter. To obtain the final high-resolution HSI $\hat{X}$, the trained SpeUnet is applied to the original MSI Z, resulting in

$$\hat{X} = \text{SpeUnet}\,(Z)\,. \tag{35}$$

This multistage loss formulation enforces consistency across augmentation, downsampling, and reconstruction, effectively improving overall image quality and supporting accurate HSI restoration.

IV. DATA AND EXPERIMENTS

A. Experimental Settings

1) Dataset Description: This study conducts a comprehensive evaluation of AEWFNet's performance across various scenarios using four different datasets: Chikusei, Houston, PaviaC, and PaviaU. These datasets vary in spatial and spectral resolutions, allowing for thorough validation of the fusion algorithm under different data distribution conditions. The Chikusei dataset [50], acquired on July 29, 2014, using the Headwall Hyperspec-VNIR-C sensor, contains 19 land cover categories. The Houston18 dataset [31], collected using the ITRES CASI 1500 sensor, was released for the 2018 IEEE GRSS Data Fusion Competition and includes two identified land cover categories. The PaviaC [51] and PaviaU [52] datasets were collected in northern Italy using the ROSIS sensor and were carefully preprocessed to remove certain spectral bands; each comprises nine land cover categories. Experimental analysis on these datasets enables an effective assessment of AEWFNet's applicability under various environmental conditions. Detailed dataset parameters are presented in Table I.

2) Experimental Details: The experimental setup of this study is based on a Python 3.10 environment and PyTorch 1.8, supported by an NVIDIA 4060Ti GPU with 16 GB of memory. The number of training steps is set to 40 000, and AdamW is used as the optimizer with a learning rate of 0.0001. To simulate different image resolutions and investigate the model's performance across various scales, spatial scale factors are set to 4, 8, and 16. Following the Wald protocol [53], simulated LR-HSI and HR-MSI are generated as inputs to the model. To reasonably simulate the spatial degradation process, an isotropic Gaussian convolution is applied to each nonoverlapping $r \times r$ pixel block in the reference image, with both the kernel size and full width at half maximum (FWHM) set to r, followed by a downsampling operation. This process generates LR-HSI according to the spatial scaling factor $r = W/w = H/h$, effectively preserving the spatial blurring and dimension reduction characteristics observed in real imaging scenarios. The input HR-MSI is simulated based on the spectral response of the WorldView-2 multispectral sensor.

3) Evaluation Metrics: This study adopts peak signal-to-noise ratio [54] (PSNR), spectral angle mapper [55] (SAM), Erreur Relative Globale Adimensionnelle de Synthése [56] (ERGAS), and root mean square error [57] (RMSE) as evaluation metrics to comprehensively assess the fusion quality of various algorithms. A thorough analysis is conducted on

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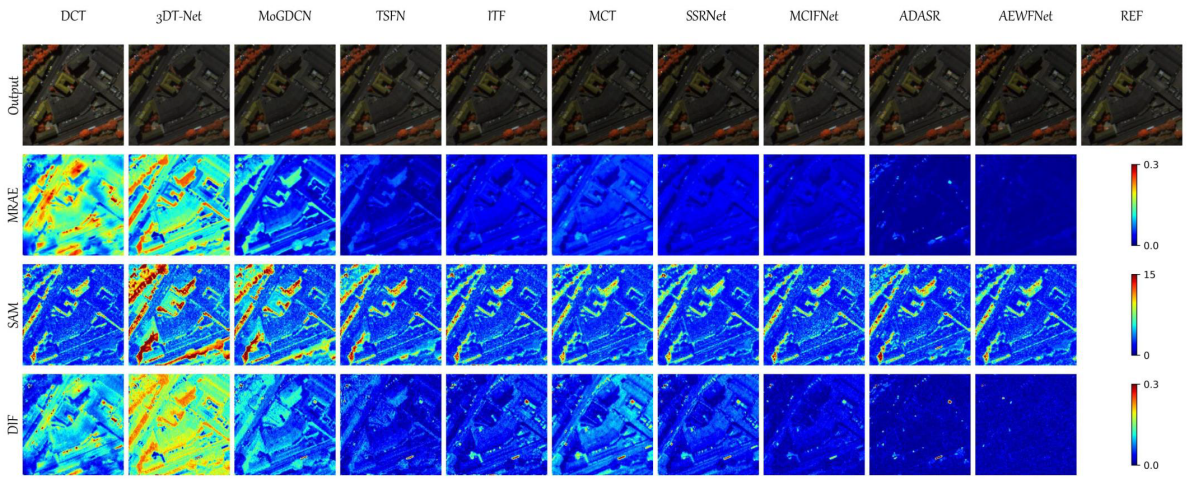


Fig. 3. Experimental results on the PaviaC dataset (R–G–B channels correspond to bands 90, 70, and 30, respectively. DIF indicates the difference image in band 25).

TABLE II
RESULTS ON THE PAVIAC DATASET

Ratio	Metrics	DCT	3DT-Net	MoGDCN	TSFN	ITF	MCT	SSRNet	MCIFNet	ADASR	AEWFNet
×4	PSNR↑	38.2097	41.3800	40.6320	41.8396	41.8191	39.8797	40.4135	42.3575	42.1951	**44.5444**
	SAM↓	3.4222	3.5269	3.6742	3.2832	3.2937	3.4300	3.3069	3.3210	3.4260	**3.2721**
	ERGAS↓	2.6441	2.0171	2.1858	1.9175	1.9236	2.2832	2.1683	1.8809	2.0217	**1.8639**
	RMSE↓	3.1337	2.1754	2.3710	2.0633	2.0682	2.5856	2.4314	1.9439	1.4714	**1.3307**
×8	PSNR↑	35.5868	41.0399	40.8654	41.1981	40.7280	39.5189	40.2834	42.4153	39.4196	**44.5093**
	SAM↓	3.4947	3.5774	3.6581	3.4340	3.4982	3.5937	3.3603	3.3410	3.5952	**3.2823**
	ERGAS↓	3.4318	2.0956	2.1370	2.0403	2.1208	2.3777	2.2089	1.8798	1.2886	**0.9350**
	RMSE↓	4.2384	2.2623	2.3084	2.2215	2.3450	2.6952	2.4681	1.9310	2.1445	**1.3371**
×16	PSNR↑	34.2375	40.2653	40.8793	40.6463	41.2279	39.7750	40.1192	42.2871	30.9540	**44.3316**
	SAM↓	3.5207	3.6764	3.6757	3.5999	3.4878	3.5796	3.4541	**3.3152**	6.8228	3.3643
	ERGAS↓	3.9854	2.2451	2.1363	2.1477	2.0485	2.3221	2.2571	1.8987	1.8381	**0.4746**
	RMSE↓	4.9507	2.4733	2.3045	2.3671	2.2138	2.6169	2.5152	1.9597	7.0249	**1.3528**

the strengths and weaknesses of different fusion methods in terms of image quality, further validating the effectiveness of AEWFNet in the HMIF task.

B. Comparative Experimental Results and Analysis

To validate the superiority of AEWFNet, this study selects several representative methods for comparison, including Transformer-based approaches such as MCT [58], DCT [28], 3DT-Net [59], and ITF [60], as well as CNN-based methods such as SSRNet [61], TSFN [62], MoGDCN [63], ADASR [14], and MCIFNet [31]. A series of fusion experiments are conducted. Specifically, in the experimental results, the best, second-best, and third-best performances are highlighted using red, blue, and green fonts, respectively.

1) Analysis on the PaviaC Dataset: In this section, the performance of AEWFNet on the PaviaC dataset is evaluated. Table II lists the performance metrics of various algorithms on this dataset, and the results indicate that AEWFNet achieves the best overall performance. In the 4× scaling task, all methods exhibit relatively similar overall performance, indicating that under lower difficulty conditions, mainstream approaches can provide stable fusion results. However, it is noteworthy that AEWFNet achieves the best performance across all evaluation metrics, demonstrating its superiority in both subjective visual quality and spectral–spatial consistency of the fusion results. In the 8× fusion task, although AEWFNet performs best overall, MCIFNet still demonstrates a certain level of competitiveness in PSNR, SAM, and RMSE. It is worth noting that the SAM value of MCIFNet differs from that of AEWFNet by only 0.0587, indicating that MCIFNet still maintains good spectral consistency at this scale factor. In the 16× fusion task, although MCIFNet outperforms AEWFNet in the SAM metric, AEWFNet still ranks first in all other metrics. This suggests that AEWFNet is more effective at preserving spatial details and reducing reconstruction errors under high-scale factors. This advantage is attributed to AEWFNet's use of WT to decompose and reconstruct features, which effectively restores edge and texture details while reducing information loss caused by large-scale upsampling. This frequency-domain feature extraction approach has been shown to enhance the sharpness of the fused images and mitigate the performance degradation typically encountered by traditional methods in high-scale fusion tasks. Since the performance of fusion algorithms is not easily distinguishable through RGB visualization, this study further presents MRAE and SAM residual heatmaps for the 8× fusion, as well as the difference map of the 25th spectral band (Fig. 3). The results show that AEWFNet yields the smallest residuals, and its fusion outputs are closest to the

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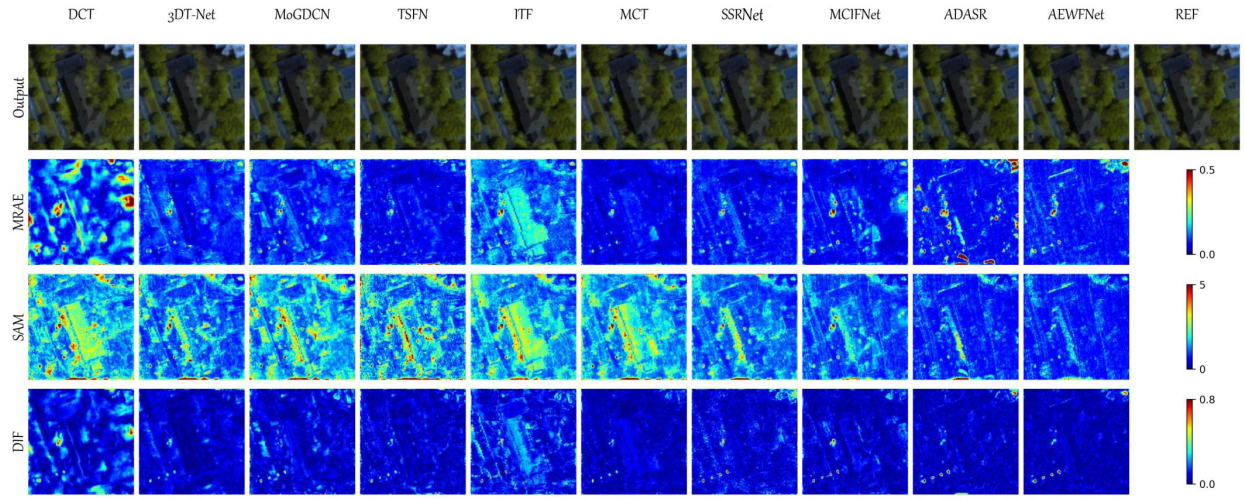


Fig. 4. Experimental results on the Houston18 dataset (R–G–B channels correspond to bands 45, 29, and 13, respectively. DIF indicates the difference image in band 25).

TABLE III
RESULTS ON THE HOUSTON18 DATASET

Ratio	Metrics	DCT	3DT-Net	MoGDCN	TSFN	ITF	MCT	SSRNet	MCIFNet	ADASR	AEWFNet
×4	PSNR↑	40.2890	46.8333	46.2881	46.2436	46.7132	46.4795	47.4303	49.1620	51.9822	**52.8802**
	SAM↓	1.5709	1.3971	1.4886	1.3578	1.4228	1.4795	1.3653	1.1254	0.9274	**0.8414**
	ERGAS↓	1.5880	0.7633	0.8140	0.8058	0.7738	0.8224	0.7201	0.5811	0.4963	**0.4336**
	RMSE↓	2.2847	1.0687	1.1452	1.1511	1.0905	1.1202	1.0041	0.8226	0.2893	**0.1950**
×8	PSNR↑	39.0515	46.2389	45.7689	44.3599	46.2363	45.5436	48.5421	49.0790	49.5190	**52.8909**
	SAM↓	1.6930	1.4134	1.5836	1.6312	1.5419	1.6378	1.1949	1.1313	0.8959	**0.8525**
	ERGAS↓	1.8205	0.8058	0.8532	0.9967	0.8220	0.8997	0.6339	0.5955	0.2443	**0.2141**
	RMSE↓	2.6345	1.1517	1.2157	1.4298	1.1520	1.2477	0.8834	0.9305	0.2294	**0.1969**
×16	PSNR↑	37.5772	45.0794	44.1250	41.9006	46.1578	45.8657	48.4875	49.0848	43.7146	**52.6808**
	SAM↓	1.7809	1.6682	1.9099	2.1346	1.5548	1.5775	1.2086	1.1362	1.5442	**0.8613**
	ERGAS↓	2.1695	0.9280	1.0595	1.3260	0.8386	0.8526	0.6360	0.5864	0.2074	**0.1063**
	RMSE↓	3.1219	1.3161	1.4690	1.8978	1.1625	1.2022	0.8890	0.8299	0.3875	**0.1974**

TABLE IV
RESULTS ON THE PAVIAU DATASET

Ratio	Metrics	DCT	3DT-Net	MoGDCN	TSFN	ITF	MCT	SSRNet	MCIFNet	ADASR	AEWFNet
×4	PSNR↑	35.8278	39.0775	37.2337	41.0163	40.2562	38.3925	38.2802	40.9749	38.4874	**41.3617**
	SAM↓	1.9820	2.0740	2.6283	1.9993	1.9590	2.1420	2.1277	1.9452	2.0989	**1.8944**
	ERGAS↓	2.3196	1.6124	1.9171	1.3266	1.4291	1.9095	1.9299	1.3209	1.7709	**1.3117**
	RMSE↓	4.1224	2.8357	3.5063	2.2684	2.4759	3.0684	3.1084	2.2793	1.9044	**1.5043**
×8	PSNR↑	31.5640	38.7244	38.1242	40.3386	40.8344	38.6676	38.0478	40.9071	37.4099	**41.2124**
	SAM↓	1.9849	2.2503	3.3872	2.2093	2.2156	1.9998	2.1569	1.9847	2.3727	**1.9747**
	ERGAS↓	3.8012	1.7366	1.8580	1.4262	1.3396	1.8509	1.9836	1.3281	0.9743	**0.6679**
	RMSE↓	6.7351	2.9534	3.1647	2.4525	2.3164	2.9728	3.1926	2.2971	2.2669	**1.5470**
×16	PSNR↑	30.0791	38.9745	38.4194	39.0393	40.0865	38.5135	37.8599	40.1564	36.3611	**40.5365**
	SAM↓	2.5450	2.2523	2.3151	2.3305	2.3439	2.3248	2.3276	3.3148	3.0043	**2.2460**
	ERGAS↓	4.5072	1.7039	1.8052	1.6972	1.4475	1.8836	2.0242	1.2909	0.5534	**0.3698**
	RMSE↓	7.9907	2.8696	3.0589	2.8482	2.5247	3.0260	3.2625	2.2321	2.7089	**1.7738**

reference image, further confirming its superiority over other algorithms.

2) Analysis on the Houston18 Dataset: Table III presents the quantitative evaluation results of different algorithms on the Houston18 dataset. AEWFNet demonstrates significant advantages across the 4×, 8×, and 16× scale factor fusion tasks. In addition, the SAM residual map in Fig. 4 shows that the images generated by AEWFNet exhibit lower overall error and a more uniform color distribution, indicating greater stability in pixel-level error control. In the 4× fusion task, AEWFNet achieves the highest ranking across all metrics, with ADASR ranking second. The difference in their rankings is minimal. This finding suggests that AEWFNet exhibits superior image reconstruction capability under low-scale factors, effectively enhancing spatial detail restoration while maintaining spectral consistency. In the 8× fusion task, the performance gap between ADASR and AEWFNet widens across all evaluation metrics, further demonstrating AEWFNet's more pronounced

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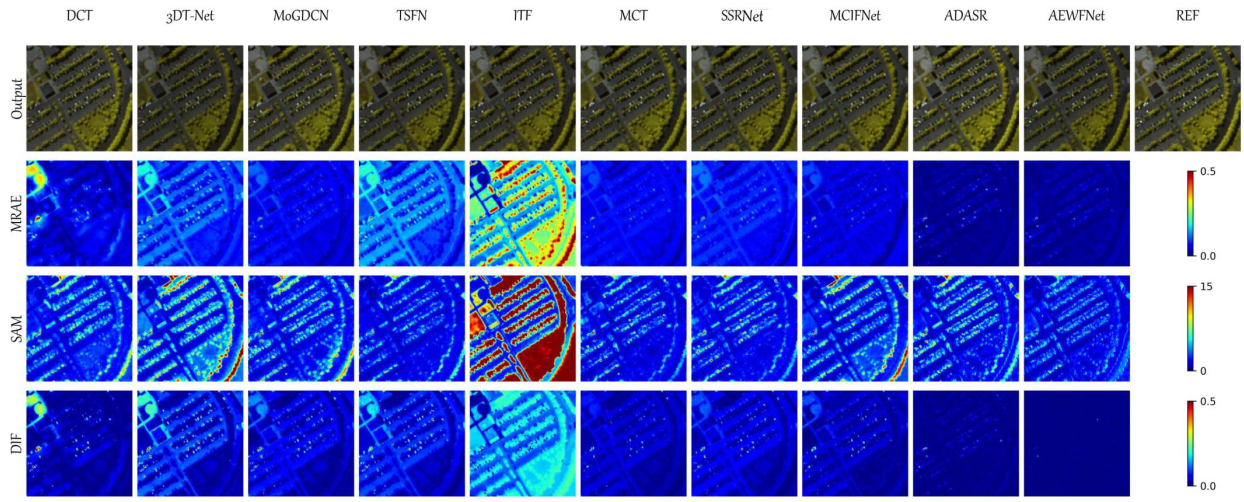


Fig. 5. Experimental results on the PaviaU dataset (R–G–B channels correspond to bands 90, 70, and 30, respectively. DIF indicates the difference image in band 25).

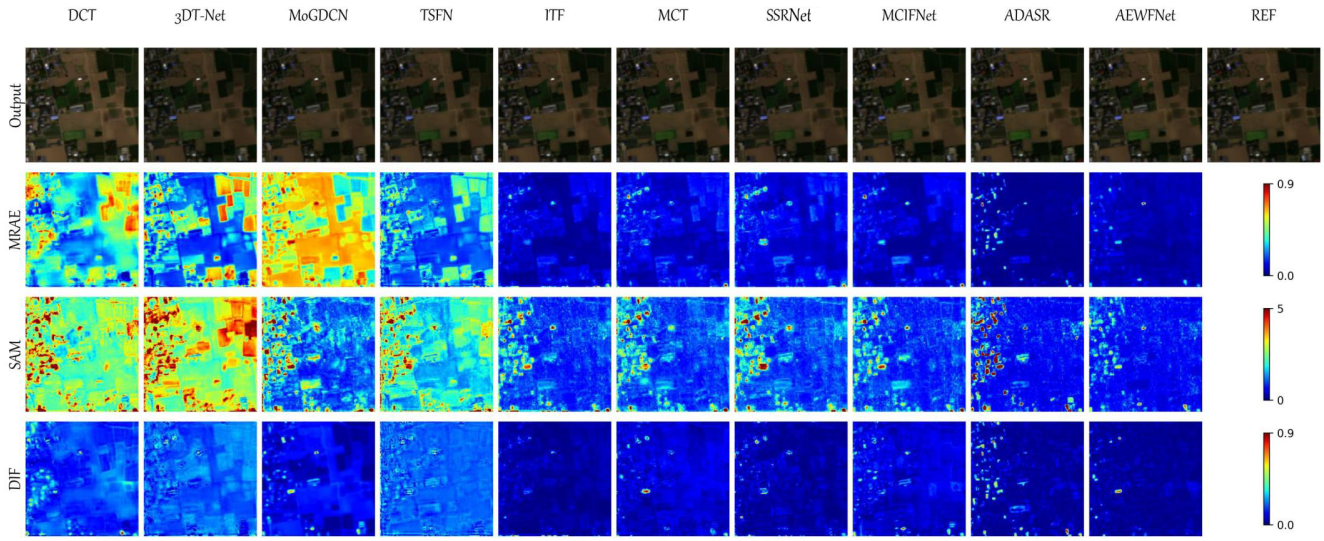


Fig. 6. Experimental results on the Chikusei dataset (R–G–B channels correspond to bands 61, 41, and 21, respectively. DIF indicates the difference image in band 25).

advantage in high-resolution image reconstruction. This trend is attributed to AEWFNet's stronger nonlinear modeling capability under high-scale factors, enabling superior performance in spectral fidelity, spatial detail preservation, and fusion error control. It is noteworthy that in the 16× fusion task, although ADASR still ranks second in ERGAS and RMSE metrics, MCIFNet has surpassed ADASR in PSNR and SAM metrics. This indicates that, in high-scale factor tasks, fusion algorithms place higher demands on spectral consistency and spatial detail recovery, and AEWFNet performs more prominently in spectral fidelity. In conclusion, the experimental results on the Houston18 dataset further demonstrate the outstanding performance of AEWFNet in the hyperspectral and multispectral fusion domain, along with its significant advantages over other deep learning models.

3) Analysis on the PaviaU Dataset: Compared to the Houston18 dataset, the PaviaU dataset has higher spectral resolution but lower spatial resolution, making the data fusion task relatively more complex. This section presents a comparative analysis of the performance of AEWFNet and other fusion algorithms on the PaviaU dataset. The results in Table IV clearly indicate that AEWFNet outperforms other methods across all evaluation metrics, further validating that the wavelet feature extraction and cross-scale feature fusion strategy adopted in this study effectively learns and preserves spectral information while reducing fusion errors in high-scale factor tasks. Additionally, in the residual maps of the 25th spectral band (as shown in Fig. 5), AEWFNet produces the smallest deviation between the fused image and the reference image, indicating that it accurately reconstructs the spectral

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TABLE V
RESULTS ON THE CHIKUSEI DATASET

Ratio	Metrics	DCT	3DT-Net	MoGDCN	TSFN	ITF	MCT	SSRNet	MCIFNet	ADASR	AEWFNet
×4	PSNR↑	38.1349	41.8523	41.9704	44.5777	44.2655	43.7090	43.5616	45.4049	**53.5277**	53.4527
	SAM↓	1.2412	1.3120	1.3595	0.9161	1.0111	1.1237	1.1216	0.9755	0.7485	**0.7342**
	ERGAS↓	1.8006	1.6540	1.5299	1.1247	1.2617	1.1159	1.2102	1.0607	1.1058	**1.0510**
	RMSE↓	2.9187	1.9025	1.8768	1.3901	1.4410	1.5363	1.5626	1.2638	0.7668	**0.7421**
×8	PSNR↑	34.4735	41.9651	41.3501	42.6289	44.0618	43.9173	43.2895	44.8715	46.5947	**53.3876**
	SAM↓	1.3854	1.3107	1.4511	1.1287	1.0578	1.0998	1.1741	1.0090	1.0476	**0.7420**
	ERGAS↓	2.4204	1.6450	1.5992	1.4221	1.2796	1.0543	1.2612	1.1773	0.7091	**0.5152**
	RMSE↓	4.4490	1.8780	2.0155	1.7398	1.4752	1.4999	1.6124	1.3439	1.1985	**0.7512**
×16	PSNR↑	31.6291	41.6825	40.8008	40.2681	44.2436	42.9001	42.9670	44.9579	42.2029	**52.3711**
	SAM↓	1.3865	1.3751	1.5732	1.4323	1.0537	1.2300	1.2069	1.0095	1.5348	**0.7771**
	ERGAS↓	2.8660	1.7378	1.7156	1.9192	1.2284	1.2485	1.2441	1.1289	0.5897	**0.2641**
	RMSE↓	6.1726	1.9400	2.1473	2.2831	1.4446	1.6863	1.6734	1.3306	1.8181	**0.7979**

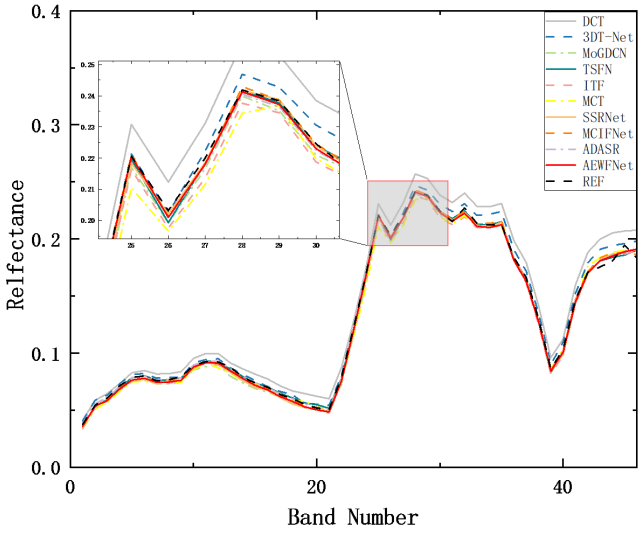


Fig. 7. Spectral curve comparison between each method and the reference ground truth on the Houston18 dataset under the 8× setting; the *x*-axis denotes the band index, and the *y*-axis denotes reflectance.

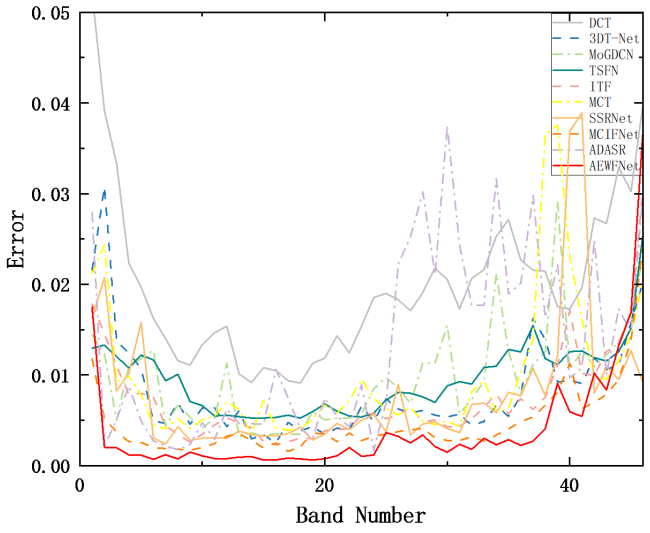


Fig. 8. Per-band error of each method with respect to the reference ground truth on the Houston18 dataset under the 8× setting; the *x*-axis denotes the band index, and the *y*-axis denotes the error value.

characteristics of HSI and maintains good generalization under complex data distributions. Although differences in the SAM heatmaps are not visually prominent, closer analysis reveals that AEWFNet achieves the smallest residuals, further confirming its effectiveness and superiority in remote sensing image fusion tasks.

4) Analysis on the Chikusei Dataset: The Chikusei dataset is characterized by relatively low spatial resolution and high spectral resolution. Due to its low spatial resolution, the dataset contains complex mixed pixels, where a single pixel may encompass multiple land cover types or visual features. Meanwhile, the high spectral resolution enables each pixel to carry richer spectral information, which enhances spectral representation capability but also increases the difficulty of information mismatch and spectral separation. Compared to the PaviaU dataset, this imposes higher demands on fusion algorithms. Table V presents the quantitative evaluation results of different fusion algorithms on the Chikusei dataset. For this data distribution, although AEWFNet exhibits a slightly lower PSNR than ADASR under the 4× scaling factor, it still achieves the best overall performance. This indicates that AEWFNet can effectively model the nonlinear mapping inherent in the fusion process for this dataset. Furthermore, the heatmap comparison shown in Fig. 6 provides additional validation: the fusion results produced by AEWFNet exhibit the lightest heatmap color among all methods, visually reflecting the smallest deviation from the reference image and thus appearing more consistent with real-world scenes. In summary, AEWFNet demonstrates superior fusion quality and robustness on the Chikusei dataset, consistently achieving accurate spectral reconstruction, effective preservation of spatial details, and significant error reduction across various scaling factors. These advantages are largely attributed to its wavelet-based feature extraction and cross-scale fusion strategies, setting it apart from other competing methods.

C. Comparison of Spectral Curves

This section analyzes the spectral curves of the fusion results on the Houston18 dataset under the 8× scale factor, evaluating the spectral fidelity of different algorithms by comparing them with the ground truth. As shown in Fig. 7,

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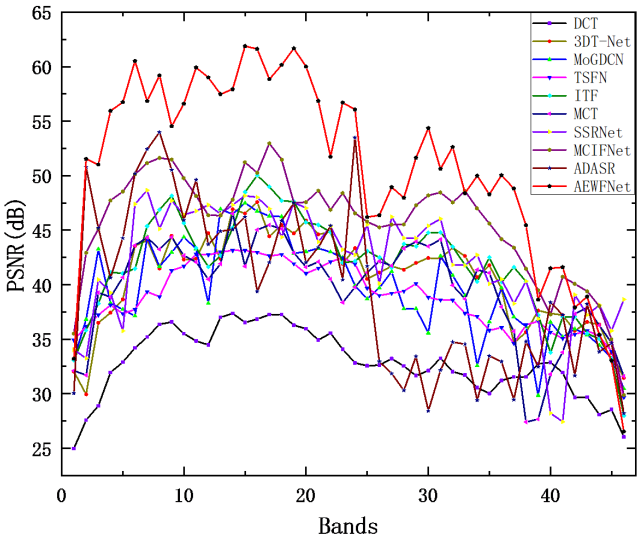


Fig. 9. Per-band PSNR statistics on the Houston18 dataset under the 8× setting; the *x*-axis denotes the band index, and the *y*-axis denotes PSNR (dB).

TABLE VI
SARL ABLATION RESULTS ON THE HOUSTON18 DATASET

	PSNR↑	SAM↓	ERGAS↓	RMSE↓
SRF	51.9649	0.8736	0.1086	0.2018
Spa Inf	52.6399	0.8629	0.1069	0.1982
SARL	**52.6808**	**0.8613**	**0.1063**	**0.1974**

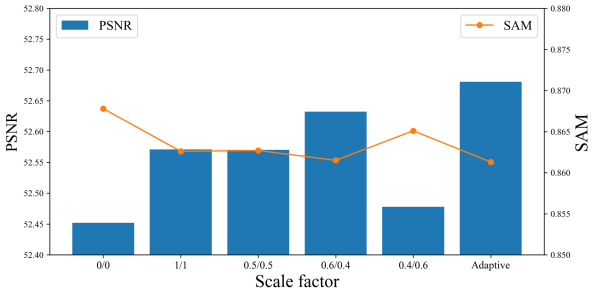


Fig. 10. Performance on the Houston18 dataset under different parameter settings.

the spectral curve generated by AEWFNet closely aligns with the real curve, and exhibits the smallest error compared to the reference curve across most spectral bands (Fig. 8), demonstrating better performance than other methods and highlighting its advantage in preserving spectral information. This result further confirms the reliability of AEWFNet in accurate spectral reconstruction, providing higher quality support for practical applications such as land cover classification and environmental monitoring that rely on high-precision spectral data.

D. Bandwise PSNR Comparison

Building upon the evaluation of overall fusion performance in the previous experiments, this section further analyzes the per-band PSNR statistics on the Houston18 dataset under the 8× scale factor to validate the advantages of AEWFNet. As shown in Fig. 9, although AEWFNet does not achieve the highest score in every individual band, it consistently demonstrates superior overall performance. This advantage is mainly attributed to the integration of wavelet convolution and cross-scale feature fusion, which enables AEWFNet to accurately capture spectral and spatial information at different scales in the frequency domain, resulting in more refined image reconstruction. In addition, AEWFNet adopts an SARL mechanism to adaptively enhance interband correlations, ensuring a high signal-to-noise ratio (SNR) across different spectral regions in the fusion results. In summary, AEWFNet exhibits outstanding performance in both overall fusion quality and per-band analysis, further confirming its effectiveness and superiority in HMIF tasks.

E. Ablation Experiment Results and Analysis

To comprehensively evaluate the contribution of each key module to the performance of AEWFNet, a series of ablation experiments are conducted on the Houston18 dataset with a fusion scale factor of 16. These experiments focus on four core components—SARL, ARPAN, SpeUnet, and the consistency loss—with the goal of analyzing the individual and combined impact of each module on the performance improvement in HMIF tasks.

1) Effectiveness of SARL: In the HMIF task, traditional spectral degradation modeling methods typically rely on fixed convolution structures, which are insufficient to flexibly adapt to the complex spectral mapping relationships between different images. To address this issue, AEWFNet introduces the SARL module, which combines the SRF with dynamic spatial information to achieve a more adaptive spectral degradation modeling mechanism. In the ablation experiments, this study replaces SARL with two alternative approaches: using only the SRF to model spectral degradation, and using only spatial information to extract spectral features. As shown in Table VI, the model incorporating the complete SARL module achieves the best performance in both fusion quality and reconstruction accuracy. These results clearly demonstrate that SARL plays a significant role in enhancing the modeling capability for hyperspectral and multispectral features. Its adaptive feature representation mechanism effectively captures complex spectral mapping relationships, thereby significantly improving fusion performance.

2) Dynamic Weighting in SARL: To further verify the importance of the dynamic coordination mechanism between SRF modeling and spatial information in SARL, this study conducts an ablation experiment focusing on the balance ratio between the two components. As shown in Fig. 10, multiple fixed ratio settings were tested, with fusion performance evaluated primarily using PSNR and SAM metrics. The results indicate that although some fixed ratios (e.g., 0.6/0.4) perform relatively well in certain metrics, they still struggle to effectively balance the integration of spatial and spectral information. In contrast, the spatially guided dynamic weighting strategy achieves the best performance in both PSNR and SAM, demonstrating stronger fusion capability and robustness.

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TABLE VII
ARPAN ABLATION RESULTS

	PSNR↑	SAM↓	ERGAS↓	RMSE↓
Random	51.8292	0.8897	0.1121	0.2037
Fixed	51.1541	0.9208	0.1247	0.2030
ARPAN	**52.6808**	**0.8613**	**0.1063**	**0.1974**

TABLE VIII
SYNERGISTIC EFFECT OF WT AND CONVOLUTIONAL LAYERS

	PSNR↑	SAM↓	ERGAS↓	RMSE↓
Convolution	52.6665	0.8647	0.1091	0.1982
WT	52.6708	0.8615	0.1072	0.1973
Synergy	**52.6808**	**0.8613**	**0.1063**	**0.1974**

TABLE IX
IMPACT OF DIFFERENT WAVELET BASES ON PERFORMANCE

	PSNR↑	SAM↓	ERGAS↓	RMSE↓
Db3	46.5952	1.4406	0.2129	0.3967
Db2	50.4203	0.9396	0.1341	0.2214
Haar	**52.6808**	**0.8613**	**0.1063**	**0.1974**

TABLE X
L_S LOSS FUNCTION RATIO

	PSNR↑	SAM↓	ERGAS↓	RMSE↓
0/0	52.4519	0.8678	0.1068	0.1990
1/1	52.5712	0.8626	0.1112	0.1984
0.5/0.5	52.5705	0.8627	0.1112	0.1984
0.6/0.4	52.6323	0.8615	0.1091	0.1978
0.4/0.6	52.4778	0.8651	0.1124	0.1990
1/0.3	**52.6808**	**0.8613**	**0.1063**	**0.1974**

This indicates that the dynamic adjustment mechanism can adaptively optimize the degree of fusion between spectral and spatial information based on the spatial distribution of input features, thereby avoiding issues of information redundancy or insufficiency. Consequently, it significantly enhances the model's ability to capture complex relationships among multisource images and improves its capability for detailed representation.

3) Impact of ARPAN: Considering the limited availability of hyperspectral data and the relatively small number of samples, models are prone to overfitting or insufficient generalization during training. To enhance the model's representation capability for spectral features, this study introduces an augmentation mechanism for the spectral outputs. Different from traditional random or rule-based fixed augmentation strategies, ARPAN adopts an adaptive augmentation approach guided by the distribution of spectral features, dynamically mining potential spectral information at the data level to enhance model robustness. As shown in Table VII, ARPAN significantly outperforms both random and fixed augmentation methods across all evaluation metrics. This clearly demonstrates that ARPAN can implement more targeted augmentation strategies based on the inherent distribution of spectral features, thereby effectively enhancing the model's capability to perceive and discriminate spectral characteristics.

4) Synergistic Effect of WT and Convolutional Layers: In the spectral upsampling network, this study designs an upsampling module that combines WT with convolutional layers to fully exploit the multiscale characteristics of wavelet decomposition and the local receptive field capability of convolution, thereby improving the quality of spectral information reconstruction. To validate the effectiveness of this module, an ablation study is conducted by comparing different configurations: WT only, convolutional layers only, and their combination. The experimental results are presented in Table VIII. While using either WT or convolution alone can improve reconstruction performance to some extent, their combination achieves the best results across all evaluation metrics. This demonstrates that the integration of WT and convolution effectively complements their respective feature extraction capabilities, enhances the model's ability to represent spectral information, reduces information loss, and improves the quality of the fused image.

5) Choice of Wavelet Basis: In the application of WT, the choice of wavelet basis has a significant impact on feature extraction and fusion performance. This study evaluates the effect of three wavelet bases—Db3, Db2, and Haar—on the fusion model, with the experimental results summarized in Table IX. As shown in the table, there are notable differences in performance across the evaluation metrics for different wavelet bases. Among them, the Haar wavelet achieves the best results in all metrics. This indicates that, compared to Db3 and Db2, the Haar wavelet is more effective in preserving spectral information integrity and provides finer feature representation in HMIF tasks.

6) L_S Loss Function: In the AEWFNet model, the L_S loss function plays a critical role during the optimization process. To investigate the impact of different weight combinations on model performance, this study conducts a series of experiments by testing various L_S loss weight settings, with the results presented in Table X. The experimental results show that different weight allocations have a certain impact on the model's performance. Among them, the model achieves the best performance across all metrics when the weight ratio is set to 1/0.3. This indicates that an appropriate weight distribution in the loss function can effectively balance the optimization between spectral and spatial information, avoiding overfitting to specific features and thereby improving the overall quality of the fused image. Therefore, this study adopts 1/0.3 as the optimal L_S loss weight configuration to further enhance the fusion performance of AEWFNet.

F. Evaluation of High-Scale Fusion Capability

Considering that the original image size leads to excessive blurring and structure loss under a scaling factor of 32, we constructed a 192×192 version of the Houston18 dataset to more reasonably evaluate the performance of our method under such extreme reconstruction conditions. This setting allows for a more rational and representative evaluation of the proposed method in extreme reconstruction scenarios. The

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TABLE XI
EXPERIMENTAL RESULTS WITH A SCALING FACTOR OF 32

Ratio	Metrics	DCT	3DT-Net	MoGDCN	TSFN	ITF	MCT	SSRNet	MCIFNet	ADASR	AEWFNet
×32	PSNR↑	26.1925	38.9080	38.4637	32.1361	34.6002	39.4783	41.9294	40.1058	41.7118	**52.0815**
	SAM↓	3.3697	3.0688	3.2230	4.0612	4.9303	3.1249	2.3077	2.9509	2.1928	**0.8946**
	ERGAS↓	7.2020	1.7835	1.9213	3.6297	2.8777	1.6790	1.4002	1.6304	1.0120	**0.4352**
	RMSE↓	12.5016	2.8916	3.0434	6.3057	4.7482	2.7079	2.0421	2.5191	1.7248	**0.6889**

TABLE XII
EXPERIMENTAL RESULTS UNDER DIFFERENT NOISE LEVELS

Ratio	Metrics	DCT	3DT-Net	MoGDCN	TSFN	ITF	MCT	SSRNet	MCIFNet	ADASR	AEWFNet
SNR=10	PSNR↑	35.2923	45.9406	44.3957	41.4095	47.2232	45.1277	46.5578	48.9866	43.4146	**52.4169**
	SAM↓	1.6271	1.4580	1.7873	1.8746	1.2726	1.5434	1.3507	1.0479	1.5416	**0.8767**
	ERGAS↓	2.3303	0.7407	0.9130	1.1792	0.6697	0.8168	0.6830	0.5374	0.2345	**0.1106**
	RMSE↓	4.3435	1.2748	1.5229	2.1477	1.0998	1.3998	1.1873	0.8977	0.4225	**0.1992**
SNR=20	PSNR↑	36.4680	45.5937	44.7483	41.5071	47.0882	46.0438	46.3245	48.7618	44.2480	**52.5765**
	SAM↓	1.5124	1.4757	1.7199	1.8639	1.3081	1.4599	1.3932	1.0900	1.5059	**0.8715**
	ERGAS↓	2.0397	0.7657	0.8681	1.1598	0.6943	0.7427	0.6982	0.5486	0.1993	**0.1068**
	RMSE↓	3.7937	1.3267	1.4623	2.1238	1.1170	1.2597	1.2197	0.9212	0.3601	**0.1992**
SNR=30	PSNR↑	36.1601	45.2725	44.3517	41.3026	47.7845	46.6929	46.6298	48.8834	44.4064	**52.6782**
	SAM↓	1.5531	1.5621	1.7926	1.8590	1.2224	1.3834	1.3479	1.0722	1.4717	**0.8670**
	ERGAS↓	2.0992	0.7963	0.9155	1.2010	0.6183	0.7006	0.6728	0.5429	0.1974	**0.1062**
	RMSE↓	3.9306	1.3767	1.5307	2.1743	1.0309	1.1690	1.1775	0.9084	0.3592	**0.1981**

experimental results demonstrate that AEWFNet outperforms competing methods across all four mainstream evaluation metrics, exhibiting superior performance in large-scale fusion scenarios. As shown in Table XI, AEWFNet achieves a significantly higher PSNR than suboptimal methods such as ADASR and SSRNet, indicating its greater capacity for high-fidelity image reconstruction. AEWFNet also achieves the best performance in terms of SAM, ERGAS and RMSE, clearly reflecting its advantages in spectral consistency, reconstruction accuracy, and error suppression. These findings further validate the effectiveness and synergy of the proposed SARL, ARPAN, and wavelet convolution strategies in handling complex degradation modeling and cross-modal information fusion.

G. Robustness Experiments

In image fusion tasks, spectral data are often affected by noise. Therefore, this study introduces noise into the test images to further evaluate the robustness of the proposed method. In the 16× scale factor fusion task on the Houston18 dataset, Gaussian noise with SNRs of 10, 20, and 30 dB is added. For this dataset, evaluation metrics are computed under different noise levels, and the average results are reported in Table XII. The experimental results clearly show that under SNRs of 10, 20, and 30 dB, the proposed AEWFNet consistently demonstrates superior fusion performance, confirming its robustness and effectiveness in noisy environments.

H. Comparison of Model Parameters

This section presents a statistical analysis of the model size of deep learning-based algorithms, as shown in Fig. 11. The horizontal axis represents the deep learning models, while the vertical axis indicates the corresponding model size, referring to the storage space occupied by each model. The size of each bubble denotes the number of model parameters. The experimental results show that among all the compared

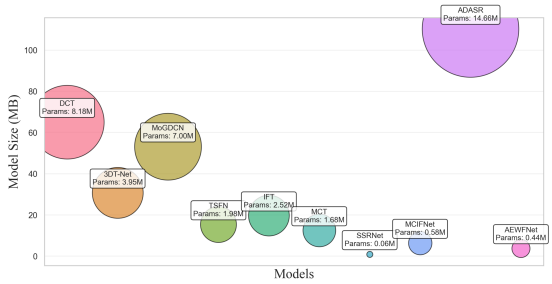


Fig. 11. Complexity statistics.

methods, AEWFNet ranks second in model size, with only a slight difference from the top-ranked SSRNet. Combined with the previous experimental findings, this result further confirms that AEWFNet achieves a well-balanced trade-off between fusion accuracy and lightweight design.

V. CONCLUSION

In order to preserve the spectral priors and multiscale structural details of the input image, while also addressing the limitations of existing methods in modeling the imaging physical mechanism, coping with the scarcity of hyperspectral data, and facilitating cross-modal feature interaction, this article proposes an HMIF framework. This framework is named AEWFNet, and it is based on adaptive enhancement and wavelet convolution. Specifically, AEWFNet introduces a SARL mechanism to assist the spectral degradation process from HR-HSI to MSI using spatial contextual cues, thereby effectively alleviating spectral distortion and information loss. Furthermore, the ARPAN module adaptively predicts rotation angles based on image content to enhance image representation, improving image quality while preserving crucial spectral and structural features. Finally, wavelet convolutions

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are employed in AEWFNet to extract multiscale frequency-domain features. This significantly boosts the spatial structure representation and edge detail fidelity of the fused image, and enables high-quality cross-modal fusion and accurate reconstruction. Despite the fact that AEWFNet evinces superior performance in terms of spectral fidelity and detail preservation, the model remains encumbered by a relatively large number of parameters. Subsequent endeavors will concentrate on the implementation of lightweight architectures or the sharing of parameters, with the objective of achieving a further reduction in model complexity while ensuring the preservation of fusion performance.

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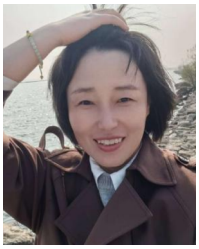
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